

Speech-Based Computational Analysis of Striatal and Basal Ganglia Dysfunction in Parkinson's Disease Using Classical and Hybrid Quantum Machine Learning

Dr. Siddique Ibrahim S P¹, Karthikeyan Bharadwaj S¹, Naga Lahari T¹, Sai Kushal Vittanala¹, and Aravind Kumar M¹

School of Computer Science and Engineering, VIT-AP University,
Amaravati, Andhra Pradesh, India
siddique.ibrahim@vitapstudent.ac.in
karthikeyan.22bce9112@vitapstudent.ac.in
lahari.22bce7250@vitapstudent.ac.in
kushal.22bce9529@vitapstudent.ac.in
aravind.22bce20503@vitapstudent.ac.in

Abstract. Parkinson's disease (PD) is a progressive neurodegenerative disorder caused by dopamine depletion in the striatum, which disrupts basal ganglia–thalamocortical circuits responsible for motor timing, force regulation, and coordinated movement. Because speech production depends on precise motor control, early striatal dysfunction manifests as measurable acoustic abnormalities, making speech a reliable and non-invasive biomarker for PD detection. In this study, we propose a comprehensive speech-based computational framework to analyze basal ganglia dysfunction using both machine learning and hybrid quantum machine learning models. A total of 754 acoustic features were extracted from sustained phonation recordings to capture perturbation measures, non-linear dynamics, spectral variations, and temporal instability associated with hypokinetic dysarthria. Machine learning models including Logistic Regression, Support Vector Machine, and XGBoost were evaluated using stratified ten-fold cross-validation. To ensure a fair and controlled comparison with quantum models under current qubit limitations, Principal Component Analysis (PCA) was applied to reduce feature dimensionality. To enable a direct and controlled comparison between quantum and machine learning models on identical input representations, PCA-reduced features were also evaluated using machine learning models. The findings confirm that high-dimensional acoustic speech features effectively reflect striatal motor control deficits and demonstrate that, under present technological conditions, machine learning provides better practical performance while quantum machine learning remains a promising exploratory direction for future neurological assessment systems.

Keywords: Parkinson's disease · basal ganglia · striatal dysfunction · speech biomarkers · machine learning · quantum machine learning

1 Introduction

Parkinson's disease is one of the most prevalent neurodegenerative disorders worldwide, affecting approximately one to two percent of individuals over the age of sixty-five.

The disease arises from progressive loss of dopaminergic neurons in the substantia nigra pars compacta, which results in dopamine depletion within the striatum. The striatum serves as the primary input nucleus of the basal ganglia and plays a central role in regulating motor timing, amplitude scaling, and movement coordination. Dopamine depletion disrupts the balance between the direct and indirect pathways of the basal ganglia, leading to impaired motor control and the characteristic symptoms of Parkinson's disease.

While Parkinson's disease is traditionally diagnosed based on motor symptoms, the disease begins with non-motor symptoms which develop in the beginning stages and continue to progress without detection. Hypokinetic dysarthria occurs in approximately seventy to ninety percent of people with Parkinson's disease and shows five distinct vocal characteristics which include lower vocal loudness and unchanging pitch and unclear speech sounds and breathy voice quality and vocal fold vibration that shows greater instability. The speech disorders result from damaged neural pathways which control speech production instead of physical harm to the vocal cords. The speech impairment stands as one of the most frequently occurring clinical disabilities.

Human beings create spoken language through an intricate motor system that requires all three components of speech production to function simultaneously. This system uses striatal-basal ganglia-cortical loops to control its functions which guarantee exact timing and correct power delivery. Scientists can measure the first signs of striatal dysfunction which cause detectable changes in speech acoustics. These changes can be quantified using digital signal processing techniques and analyzed using machine learning methods.

The figure shows the connection between the basal ganglia and the striatum in Parkinson's disease and how their dysfunction affects speech. The basal ganglia are linked to features such as DFA (Detrended Fluctuation Analysis), RPDE (Recurrence Period Density Entropy), and PPE (Pitch Period Entropy), while the striatum influences acoustic measures like jitter, shimmer, and HNR (Harmonics-to-Noise Ratio). The overlap represents common speech abnormalities such as voice instability and rhythm irregularity.

Recent advances in machine learning have created automated systems which analyze speech signals to detect Parkinson's disease. Most research studies assess predictive accuracy as their main objective but their results do not deliver sufficient understanding of neurophysiological processes. The evaluation of new quantum machine learning techniques lacks proper experimental controls and their performance assessment remains unclear when compared to conventional methods. The researchers developed a new evaluation framework for quantum machine learning models which requires both classical and hybrid models to be tested with the same set of features. The primary contribution of this work lies in

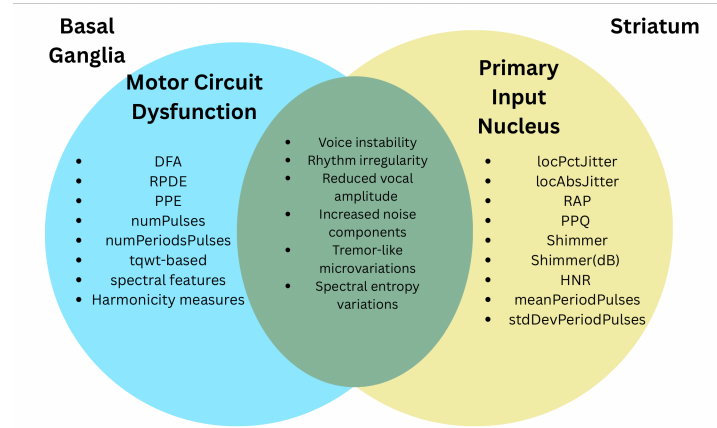


Fig. 1. Venn diagram illustrating shared and distinct speech biomarkers associated with basal ganglia and striatal dysfunction in Parkinson's disease.

providing an honest comparative analysis while explicitly linking acoustic speech features to striatal and basal ganglia dysfunction.

2 Literature Review

Approximately 70 to 90 percent of patients with Parkinson's disease experience speech problems which represent the most common and medically important symptoms of their condition. The primary vocal characteristics of acoustic abnormalities include diminished vocal strength and unchanging pitch and breathiness together with unclear speech production. Vasquez-Correa et al. [5] conducted a detailed survey of speech biomarkers used in PD research while they demonstrated that jitter and shimmer together with harmonic-to-noise ratio (HNR) measurements serve as effective indicators for assessing motor timing instability and force control deficits. Tsanas et al. [6] showed that doctors can use remote speech analysis to track disease development over time because this method detects even minor problems with vocal fold control through sustained phonation tasks. Xu et al. [7] discovered that machine learning models which learn from acoustic biomarkers reached area under the curve (AUC) scores above 0.90 in all standard test datasets, which demonstrates that speech functions as a trustworthy noninvasive method for diagnosing medical conditions. Ji et al. [9] demonstrated how speech biomarkers behave differently in various subtypes because different PD phenotypes display distinct patterns of striatal dysfunction. Valarmathi et al. [10] demonstrated how acoustic perturbation features link to striatal dopamine loss because speech disorders show basic vocal modifications which result from basal ganglia motor circuit breakdown.

High-dimensional acoustic features have become more accessible which creates an opportunity for classical machine learning methods to dominate Parkinson’s disease speech analysis. Xu et al. [7] showed that Support Vector Machines and Random Forests together with gradient boosting methods achieve strong classification accuracy when they use complete acoustic feature collections for training. The study demonstrated that ensemble-based models successfully managed two speech dataset challenges which involved handling correlated perturbation together with nonlinear speech characteristics. Researchers have developed deep learning methods to discover unique representation patterns through automatic learning. Sedigh Malekroodi et al. [8] used transformer-based speech representation learning to achieve better classification results than their hand-crafted feature models which showed how pre-trained embeddings can effectively model complex temporal and spectral relationships. The models need extensive datasets together with heavy computational power which creates a barrier for their use in medical settings. Simone et al. [11] showed that explainable artificial intelligence (XAI) plays a vital role in neurological evaluation because black-box deep learning systems decrease both clinical understanding and system reliability. Classical machine learning models maintain their competitive edge in structured acoustic datasets because they offer better sound interpretation capabilities and need less computing power while producing consistent results.

Scientists have started to investigate quantum machine learning as a new research area which studies biomedical data in recent years. Chow [12] examined how QML can help clinical decision support systems but he showed that existing Noisy Intermediate-Scale Quantum (NISQ) systems present major obstacles because they lack sufficient qubits and their circuits can only operate at limited depths. The research conducted by Rahman and Zhuang [13] focused on developing noise-aware quantum neural networks that classify biomedical signals through their studies which found that quantum circuits successfully capture nonlinear relationships but their performance depends on both feature dimensionality and optimization methods. The research conducted by Demidik et al. [14] studied quantum kernel methods which used real-world conditions to demonstrate that aggressive dimensionality reduction methods usually resulted in decreased performance. The study conducted by Pandey and Mandal [15] assessed hybrid quantum-classical learning systems and they found that fair testing of these systems against classical systems requires both identical feature representation and strict control of testing procedures. Current evidence fails to show that quantum methods provide consistent advantages in medical speech classification according to their theoretical benefits.

3 Methodology

3.1 Data Collection

The study employs speech recordings which were collected from people who have Parkinson’s disease and from healthy control participants. The researchers used

sustained vowel phonation tasks because they help identify minor vocal fold vibration problems which motor control scientists can assess through striatal and basal ganglia examination. Each speech sample is represented by a high-dimensional numerical feature vector which contains 755 attributes including a binary class label that shows whether the disease is present or absent. The dataset shows class imbalance because Parkinson’s disease samples outnumber control samples. To preserve the natural distribution of the data and avoid introducing artificial bias, stratified sampling was applied during train–test splitting and cross-validation procedures.

3.2 Data Preprocessing

The researchers conducted extensive data preprocessing work to achieve two objectives, which were to confirm data quality and maintain statistical validity, before they started model training. The researchers first converted all features into numerical format because machine learning algorithms needed this step to proceed. The dataset contained missing values, which affected the nonlinear features and higher-order acoustic features. The missing values were filled through median imputation, which proved effective because it handles outlier conditions that typically occur in speech data.

The researchers applied feature scaling techniques to achieve uniformity across the diverse range of feature values. The researchers used standardization to convert features into a distribution that has zero mean and unit variance, which is essential for both optimization-based and distance-based algorithms used in classical machine learning models. Min-Max normalization, which bounded features into an appropriate range for quantum angle encoding, served as the feature mapping technique for quantum machine learning models. All preprocessing steps were fitted exclusively on training data within each fold to prevent information leakage and ensure unbiased evaluation.

3.3 Feature Representation and Dimensionality Reduction

We extracted 754 different acoustic speech features which enabled them to study multiple aspects of speech motor control that Parkinson’s disease impacts. The features include jitter and shimmer perturbation measurements together with harmonic-to-noise ratio metrics and nonlinear dynamic features and spectral descriptors and temporal energy-based measures. The high-dimensional representation delivers substantial data about striatal and basal ganglia dysfunction but it creates problems for both computational efficiency and quantum model implementation.

Principal Component Analysis (PCA) functioned as the solution to shrink the feature space through its dimensionality reduction capability. PCA created a lower-dimensional space from the original feature set while preserving all important variance information. The researchers chose four principal components which demonstrated genuine operational limits of existing quantum computing systems because each feature dimension equates to one qubit. Importantly, PCA

was applied only on training data within each validation fold and then used to transform the test data, maintaining strict separation between training and evaluation.

3.4 Model Selection and Training

Following preprocessing and feature representation, both classical and hybrid quantum machine learning models were trained and evaluated. Classical models were first trained on the complete high-dimensional feature space to establish a strong baseline.

Logistic Regression. In this study, Logistic Regression is used as a linear classification model to distinguish Parkinson’s disease speech samples from healthy controls based on extracted acoustic features. The model learns a linear decision function over the feature space.

The decision function is defined as:

$$f(X) = \beta_0 + \sum_{i=1}^n \beta_i X_i \quad (1)$$

Where X_i represents the extracted speech features, and β_i are the learned model coefficients. The sign of $f(X)$ determines the class label, separating Parkinson’s disease and healthy control speech samples.

Support Vector Machine (SVM). Support Vector Machine is employed to identify an optimal separating hyperplane that maximizes the margin between Parkinson’s disease and healthy speech samples in the feature space.

The optimization objective is expressed as:

$$\min \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \xi_i \quad (2)$$

Where w defines the hyperplane, C is a regularization parameter, and ξ_i are slack variables allowing margin violations.

Extreme Gradient Boosting (XGBoost). XGBoost is an ensemble learning approach that builds multiple decision trees sequentially, where each new tree corrects the errors of the previous ones.

The objective function is given by:

$$\mathcal{L} = \sum_{i=1}^N l(y_i, \hat{y}_i) + \Omega \quad (3)$$

Where $l(\cdot)$ denotes the classification loss, and Ω represents a regularization term controlling model complexity.

These models were trained using stratified ten-fold cross-validation to ensure robust performance estimation. To enable a fair comparison with quantum machine learning models, the same classical models were retrained using the PCA-reduced feature space. This ablation-based approach ensures that performance differences between classical and quantum models are not attributed to unequal feature representations.

Quantum Feature Encoding. For hybrid quantum machine learning models, the reduced speech feature vector is encoded into quantum states using angle encoding, enabling representation in a quantum feature space.

The encoding is defined as:

$$|\psi(x)\rangle = \bigotimes_{i=1}^d R_y(x_i)|0\rangle \quad (4)$$

Where d is the number of encoded features, and $R_y(x_i)$ is a rotation gate parameterized by feature x_i .

Quantum Support Vector Machine (QSVM). In Quantum Support Vector Machine, a quantum kernel is used to compute similarity between samples based on quantum state overlap.

The quantum kernel is defined as:

$$K(x, x') = |\langle \psi(x) | \psi(x') \rangle|^2 \quad (5)$$

This kernel replaces the classical kernel function used in standard SVMs.

Variational Quantum Classifier (VQC). The Variational Quantum Classifier utilizes a parameterized quantum circuit, where circuit parameters are optimized using a classical optimizer to perform classification.

The variational quantum state is expressed as:

$$|\psi(x, \theta)\rangle = U(\theta)U_\phi(x)|0\rangle \quad (6)$$

Where $U_\phi(x)$ encodes the input features, and $U(\theta)$ is the trainable variational circuit.

4 Results and Discussion

4.1 Classical Machine Learning Model Performance

The performance of classical machine learning models was evaluated using Logistic Regression, Support Vector Machine, and XGBoost on the test dataset. The evaluation metrics considered include accuracy, recall, F1-score, and area

Table 1. Performance of Classical Machine Learning Models

Model	Accuracy	Recall	F1-Score	AUC
Logistic Regression	0.8553	0.9381	0.9060	0.8483
SVM	0.8224	0.9646	0.8898	0.8745
XGBoost	0.8750	0.9735	0.9205	0.9274

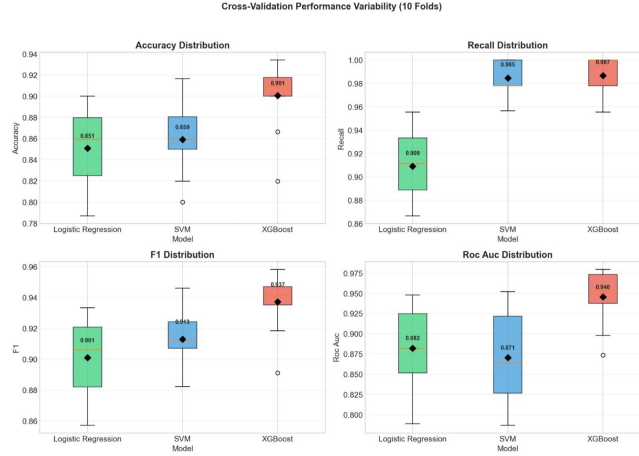


Fig. 2. Cross-validation performance variability of classical machine learning models.

under the ROC curve (AUC). The results obtained from the experiments are summarized in Table 1.

Figure 2 presents the variability of classical machine learning model performance across ten-fold cross-validation. The box plots demonstrate the stability and consistency of accuracy, recall, F1-score, and AUC across different folds.

The results indicate that all classical models achieve strong performance in detecting Parkinson’s disease from speech features. XGBoost achieves the highest accuracy of 87.5% and the highest F1-score of 92.05%, demonstrating its effectiveness in modeling complex nonlinear relationships among high-dimensional acoustic features. The ensemble nature of XGBoost allows it to handle correlated features such as jitter, shimmer, and nonlinear measures more effectively than linear models.

Figure 3 shows the ROC curves of classical machine learning models, illustrating their discriminative ability across different classification thresholds. The curves highlight the superior performance of XGBoost compared to Logistic Regression and SVM.

Support Vector Machine achieves the highest recall value of 96.46%, indicating strong sensitivity in identifying Parkinson’s disease cases. This behavior is particularly desirable in clinical screening scenarios, where minimizing false negatives is critical. However, its slightly lower accuracy suggests that this high sensitivity comes at the cost of increased false positives. Logistic Regression

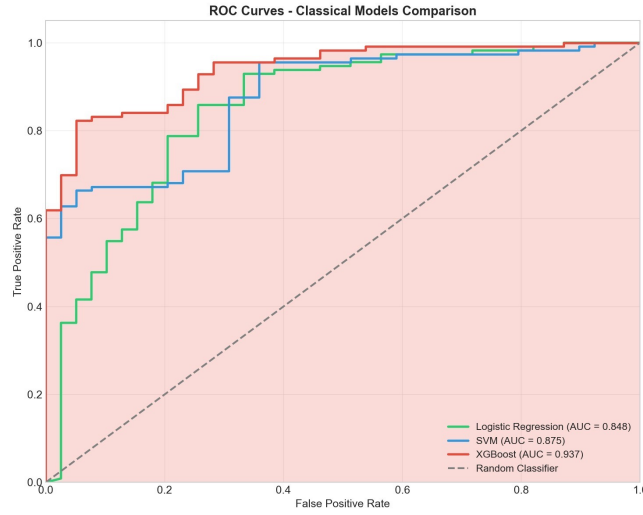


Fig. 3. ROC curves of classical machine learning models.

provides a strong baseline with balanced performance, confirming that speech features carry statistically meaningful information related to disease status.

Overall, the classical machine learning results demonstrate that speech-based acoustic features are highly effective for Parkinson’s disease detection and that ensemble and nonlinear models offer performance advantages when sufficient feature information is available.

4.2 Quantum Machine Learning Model Performance

Hybrid quantum machine learning models were evaluated using Quantum Support Vector Machine (QSVM) and Variational Quantum Classifier (VQC/QNN). Due to current limitations in quantum hardware and simulation environments, models were trained on PCA-reduced feature representations. The performance results are summarized in Table 2.

Table 2. Performance of Quantum Machine Learning Models

Model	Accuracy	Recall	F1-Score	AUC
QSVM	0.7566	0.9735	0.8560	0.73
VQC / QNN	0.7434	0.9735	0.8494	0.76

Tables 2 and 3 present a comparative analysis of quantum and classical machine learning models using PCA-based reduced speech features for Parkinson’s disease detection. The quantum models (QSVM and VQC) exhibit exceptionally

Table 3. Performance of Classical Machine Learning Models with PCA

Model	Accuracy	Recall	F1-Score	AUC-ROC
Logistic Regression	0.8146	0.9468	0.8844	0.8018
SVM	0.8162	0.9600	0.8866	0.7836
XGBoost	0.7966	0.9114	0.8698	0.8192

high recall values of 97.35%, indicating strong sensitivity in correctly identifying Parkinson’s disease cases and suggesting that quantum feature mapping effectively captures core disease-related patterns within the reduced feature space. In contrast, the classical models demonstrate a more balanced overall performance profile: Logistic Regression achieved the highest F1-score (0.8844), reflecting a strong trade-off between precision and recall; SVM attained the highest recall (0.9608) among classical approaches; and XGBoost recorded the highest AUC-ROC (0.8192), indicating superior discriminative capability. Overall, while quantum models emphasize sensitivity, classical models provide a more balanced performance across multiple evaluation metrics, highlighting complementary strengths between the two approaches.

However, quantum models exhibit lower overall accuracy and F1-score compared to classical models. QSVM achieves an accuracy of approximately 75.66%, while VQC achieves approximately 74.34%. The reduced accuracy is primarily attributed to lower specificity, indicating difficulty in distinguishing healthy control samples under aggressive dimensionality constraints. The approximate AUC values further confirm moderate discriminative capability, with VQC showing slightly higher AUC than QSVM.

These results reflect current limitations of quantum machine learning, including restricted feature dimensionality, shallow circuit depth, and optimization challenges, rather than shortcomings of the speech features themselves.

4.3 Comparative Discussion and Clinical Interpretation

The research demonstrated distinct and understandable patterns when comparing classical machine learning methods with quantum machine learning methods. The research showed that classical models achieved better accuracy and F1-score results when trained on all available features, which demonstrated the necessity of detailed high-dimensional acoustic data for accurate Parkinson’s disease identification. The reduction in performance observed in quantum models is consistent with the necessity of aggressive dimensionality reduction imposed by current quantum computing constraints.

Figure 4 compares the performance of classical and quantum machine learning models using accuracy, recall, F1-score, and AUC metrics. The comparison highlights performance differences arising from dimensionality reduction and model constraints. Despite lower accuracy, quantum models exhibit recall values comparable to or even exceeding those of classical models. The study demonstrates that quantum models can accurately identify Parkinson’s disease but fail

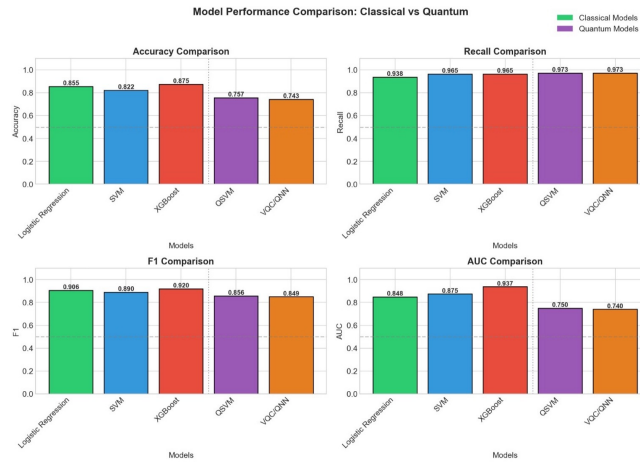


Fig. 4. Performance comparison between classical and quantum models.

to distinguish between sick and healthy speech samples. The results demonstrate that the test results are useful for initial screening tests because they effectively detect patients despite their need for further testing.

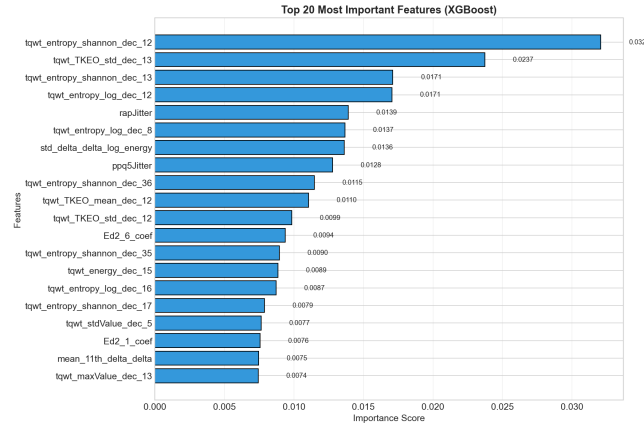


Figure 4.3 shows the top speech features identified by the XGBoost model based on importance scores. Perturbation and entropy-related features dominate, indicating strong association with striatal and basal ganglia motor dysfunction. Importantly, the separation of classical and quantum results into distinct tables allows for transparent interpretation without overstating quantum advantages. The findings demonstrate that, under current technological constraints, classical machine learning remains more suitable for practical deployment in speech-based

Parkinson's disease assessment. At the same time, quantum machine learning shows promise as an exploratory approach, warranting further investigation as quantum hardware and algorithms mature.

5 Conclusion and Future Scope

The researchers developed a speech-based computational system which enables Parkinson's disease analysis through investigation of acoustic characteristics which indicate striatal and basal ganglia dysfunction. The team extracted 754 speech features which they assessed through traditional machine learning techniques and their hybrid quantum machine learning models. The results demonstrate that speech signals contain meaningful information related to motor timing and force regulation deficits characteristic of Parkinson's disease. Classical machine learning models trained on the full feature space achieved strong performance, with ensemble-based methods such as XGBoost providing the highest accuracy and F1-score. The use of speech-based methods for disease detection demonstrates their effectiveness through high recall results which all traditional models show. The hybrid quantum models tested with PCA feature restriction achieved high sensitivity results but showed decreased accuracy because of existing limitations in their feature dimensionality and circuit expressivity and optimization capabilities. The research demonstrates through its adopted fair comparison method that the performance decline in quantum models because of aggressive dimensionality reduction also affects classical models which experience identical restrictions. Feature importance analysis confirms that perturbation-based measures such as jitter and shimmer play a dominant role, consistent with known striatal and basal ganglia motor control deficits.

Future work will focus on increasing the framework to connected and spontaneous speech, incorporating multimodal biomarkers, and applying explainable AI techniques to improve clinical interpretability. As quantum hardware matures, future studies can explore richer feature representations and deeper quantum circuits to reassess the potential of quantum machine learning in neurological assessment.

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